

Section 5.2 Independence and Dimension

Exercise 5.2.6

Use Theorem 5.2.3 to determine if the following sets of vectors are a basis of the indicated space.

b. $\{(1, 1, -1), (1, -1, 1), (0, 0, 1)\}$ in $\mathbb{R}^3$

Checking first, let us multiply $\rightarrow \det(M) = 0$? Basis! Basis.

$$\det \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ -1 & 1 & 1 \end{bmatrix} \Rightarrow 1(-1) - 1(1) + 0 = -2 \neq 0. \text{ Not Basis of } \mathbb{R}^3$$

Same thing here

w/ $\mathbb{R}_2$ have 2 of, pick that row

$$A = \begin{bmatrix} 2 & 1 & 0 & -1 \\ 1 & 1 & 0 & 2 \\ -1 & 0 & 0 & 3 \\ 3 & 2 & -3 & 1 \end{bmatrix} \Rightarrow \det(A) = -\det \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & -3 \\ -1 & 2 & 1 \end{pmatrix} - 3 \det \begin{pmatrix} 1 & 1 & 2 \\ 1 & 1 & 2 \\ 0 & 1 & -3 \end{pmatrix}$$

$$\begin{aligned} & \textcircled{1} \quad 1(1+1) - 1(0-3) + 2(1) \\ & \quad 1(1(1)-(-3)(2)) = 1+6=7 \\ & \quad 1(1(-3)-2(1)) = 5 \\ & \quad 7+5=12 \end{aligned}$$

$$\begin{aligned} & \Rightarrow \textcircled{1} + \textcircled{2} = 7 \\ & \quad 2(1(3)-2) = -10 \\ & \quad -1(-3)-3 = 6 \\ & \quad -4 \end{aligned}$$

$$\begin{aligned} & \textcircled{1} \quad 1(1+1) - 1(0-3) + 2(1) \\ & \quad 1(1(1)-(-3)(2)) = 1+6=7 \\ & \quad 1(1(-3)-2(1)) = 5 \\ & \quad 7+5=12 \end{aligned}$$

Exercise 5.2.7

In each case show that the statement is true or give an example showing that it is false.

- a. If $\{x, y\}$ is independent, then $\{x, y, x + y\}$ is independent.

Take as now the (2) axis is dependent on (1) axis.
But still linear!

- c. If $\{y, z\}$ is dependent, then $\{x, y, z\}$ is dependent for any x .

True! x is linearly dependent by $ax + bz = 0$

Then $ax + bz + cz = 0$!
where $a, b \neq 0$! the
set is dependent regardless
of x

(f) If $ax + by + cz = 0$, then $\{x, y, z\}$ is independent.

Take, as an with last question (same setup):

For given set has more than solution for $a=b, c=0, b=a$
proving 2 solutions

k. Every set containing a spanning set is again a spanning set.

True!! Take ~~where V is a spanned~~ V is a spanned: This

But V subset of spanning set
are NOT spanning set!

Let V be vector space of S , if holds V in V can be written
as a linear combination of S , adding more vector that are
also linear combination of S doesn't change anything since
they're all essentially the same vector in a different form.

Exercise 5.2.10

If $\{x_1, x_2, x_3, x_4, x_5, x_6\}$ is an independent set of vectors, show that the subset $\{x_2, x_3, x_5\}$ is also independent.

'A'

Assuming Group 'A' has all independent sub-groups 'ab' not

it holds that any normalized mode with elements in 'A' must also independent for each element being used once.



GR (non-group theory) could say $\sum x_i a_i + \dots + x_n a_n = 0$, in
then coeff must be 0, with subset all coefficients 0 for solution!

Exercise 5.2.20

Let U and W denote subspaces of $\mathbb{R}^n$, and assume that $U \subseteq W$. If $\dim W = 1$, show that either $U = \{0\}$ or $U = W$.

$$\downarrow \because \dim U \leq 1$$

Assuming no fractional dimensions (irrelevant linearly),
 By definition of 'C', if $W = \bigcirc_W$, $U = \bigcirc_W$ or $U = \bigcirc_{a,W}$

as its essentially saying $U \subseteq W$ hence U may only be 0 , 1 for dimension.

Section 5.4 Rank of a Matrix

Exercise 5.4.1

In each case find bases for the row and column spaces of A and determine the rank of A .

a. $A = \begin{bmatrix} 2 & -4 & 6 & 8 \\ 2 & -1 & 3 & 2 \\ 4 & -5 & 9 & 10 \\ 0 & -1 & 1 & 2 \end{bmatrix}$

1) REF form
RANK = 1st Lt

$$B_{\text{out}} = n_{\text{in}} - 0.4 n_{\text{out}}$$

⇒ Putting into REF form, a lot of work;

$$1) R_2 = R_2 - R_1 \quad R_3 = R_3 - R_1$$

$$\Rightarrow \begin{bmatrix} 2 & -4 & 6 & 8 \\ 0 & 3 & -3 & -6 \\ 0 & 3 & -3 & -6 \\ 0 & 1 & -1 & 2 \end{bmatrix} \Rightarrow \begin{matrix} R_4 = R_4 + \frac{1}{3}R_3 \\ R_3 = R_3 - R_2 \end{matrix}$$

7 domo; $\text{Rank} = 2$
 $\text{Rank} = 2$
 $\begin{bmatrix} 1-2 & 3 & 4 \\ 0 & 1 & -1 & -2 \end{bmatrix}$

$$p. A = \begin{bmatrix} 1 & -3 & 9 & -3 \\ 2 & -6 & 3 & -2 \\ 3 & -1 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & -3 & 9 & -3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\frac{1}{3}R_2} \begin{bmatrix} 1 & -3 & 9 & -3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \text{Rank}$$

11
Plane Rank = 2

In the space, $\left. \begin{array}{l} \text{- false print color} \\ \text{- a set of netter} \end{array} \right\} \Rightarrow$

original
print color

✓

1/1

$C = \left\{ \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} \right\}$

Test

0) look at each a Row into matrix

1) REF
2) Buena

In each case find a basis of the subspace U.

Exercise 5.4.2

a. $U = \text{span}\{(1, -1, 0, 3), (2, 1, 5, 1), (4, -2, 5, 7)\}$

$$\Rightarrow \begin{bmatrix} 1 & -1 & 0 & 3 \\ 2 & 1 & 5 & 1 \\ 4 & -2 & 5 & 7 \end{bmatrix} \xrightarrow{R_2 - 2R_1, R_3 - 4R_1} \begin{bmatrix} 1 & -1 & 0 & 3 \\ 0 & 3 & 5 & -5 \\ 0 & 2 & 5 & -5 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & -1 & 0 & 3 \\ 0 & 2 & 5 & -5 \\ 0 & 3 & 5 & -5 \end{bmatrix} \xrightarrow{R_3 - R_2} \begin{bmatrix} 1 & -1 & 0 & 3 \\ 0 & 2 & 5 & -5 \\ 0 & 1 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & -1 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 2 & 5 & -5 \end{bmatrix} \xrightarrow{R_2 \cdot 2, R_3 - 2R_2} \begin{bmatrix} 1 & -1 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 5 & -5 \end{bmatrix} \xrightarrow{R_3 \cdot \frac{1}{5}} \begin{bmatrix} 1 & -1 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix} \xrightarrow{R_1 + R_2} \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

$\xrightarrow{R_3 \cdot 2} \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & -2 \end{bmatrix} \xrightarrow{R_3 \cdot \frac{1}{2}} \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}$

REF / Buena then s/t Buena =

the same

Dimension = 3

$\Rightarrow \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix} \xrightarrow{R_1 - 3R_3} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix}$

$R_1 = R_2 - R_3$

$R_2 = R_2 - R_3$

$R_3 = R_3 - 2R_1$

d. $U = \text{span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 5 \\ -6 \end{bmatrix}, \begin{bmatrix} 2 \\ 6 \\ 8 \\ -8 \end{bmatrix}, \begin{bmatrix} 3 \\ 7 \\ 10 \\ -10 \end{bmatrix}, \begin{bmatrix} 4 \\ 8 \\ 12 \\ 12 \end{bmatrix} \right\}$

same steps since Buena = REF w/ pivot

only pivot col

REF!

$\Rightarrow u = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow R_2 = R_2 - 2R_1, R_3 = R_3 - 3R_1, R_4 = R_4 - 4R_1$

$\Rightarrow R_2 = R_2 - 2R_1, R_3 = R_3 - 3R_1, R_4 = R_4 - 4R_1$

$\Rightarrow R_2 = R_2 - 2R_1, R_3 = R_3 - 3R_1, R_4 = R_4 - 4R_1$

Since col $\begin{bmatrix} 1 \\ 1 \\ 5 \\ -6 \end{bmatrix}$ is pivot col, not included as

we col₂ bc pivot

Buena is $\left\{ \begin{bmatrix} 1 \\ 1 \\ 5 \\ -6 \end{bmatrix}, \begin{bmatrix} 2 \\ 6 \\ 8 \\ -8 \end{bmatrix}, \begin{bmatrix} 3 \\ 7 \\ 10 \\ -10 \end{bmatrix}, \begin{bmatrix} 4 \\ 8 \\ 12 \\ 12 \end{bmatrix} \right\}$

and dimension = 3

2 ways if looking at pivot col?

for row lead in col, buena

= all non 0 rows

for col, all pivot col